Welcome to COGS 18: Introduction to Python

COGS 18

Shannon E. Ellis
Associate Teaching Professor
Department of Cognitive Science, HDSI
✉ sellis@ucsd.edu

hello

my name is

Shannon Ellis

Liz

Ishan

Marlyn

Reva

Sai

Zoe

TAs

The (dreaded) waitlist

1. I do not handle the waitlist - our staff (cogsadvising@ucsd.edu) do
2. I do not have access to the waitlist nor the system that enrolls students from the waitlist.
3. Typically ~1-2 students from each section are enrolled (drop-dependent).
4. Enrollment cannot exceed no. of seats due to testing limitations.
5. The waitlist clears at the end of week 2.
6. You should complete the pre-course survey if you hope to be enrolled.

If you email me about the waitlist or your specific circumstance/need to take this course this quarter, I will point you to cogsadvising@ucsd.edu.

Let's chat: Teaching & Learning Programming

Why Python?

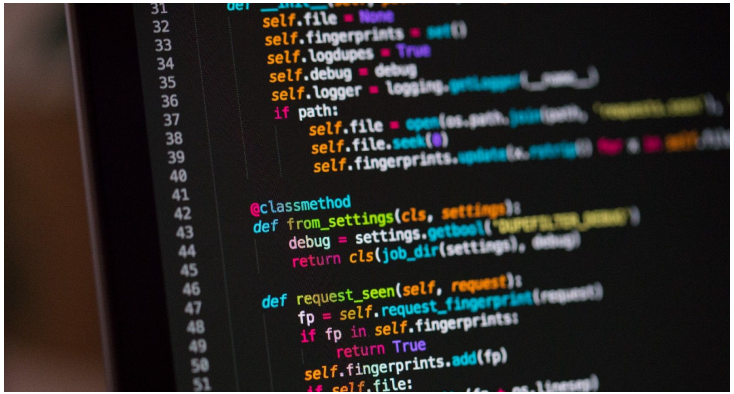
Understanding computing is critical

simple(r) syntax

widely-used

Jupyter Notebooks

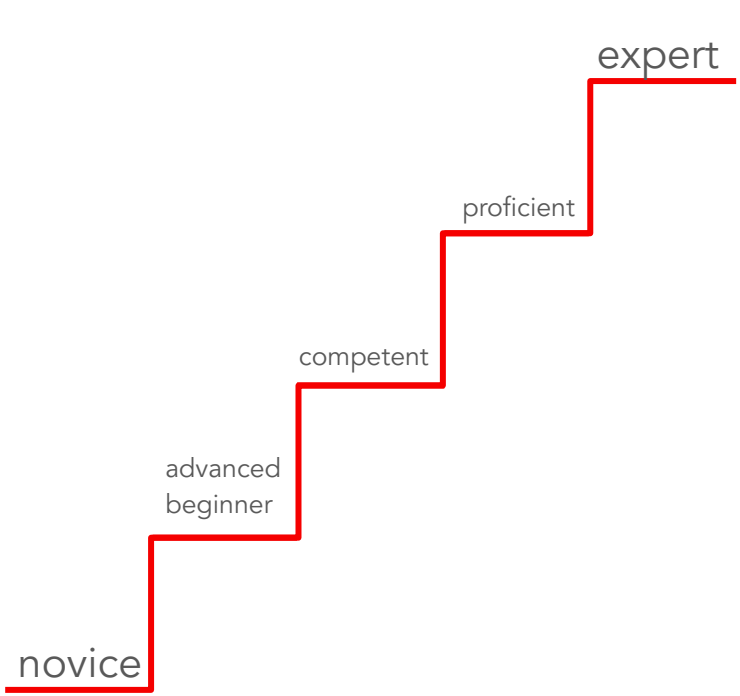
Intro Programming courses are often thought of as difficult and are courses with the highest dropout rates



….yet, the only thing that is slightly predictive of success in an intro programming course is…*how successful the student thinks they will be*

Things that do NOT predict success:

- gender
- age
- personality
- math ability



My goal is to have you all be able to program at an introductory level

It's generally accepted that it takes people 10 years to move from novice to expert programmer. But, there are lots of steps in between! We're working to move you further away from novice (& in the direction of expert) than you are right now.

Building Blocks: Too often, we also tell people to "just try things out" without explaining basic concepts. Other courses aren't taught this way...

COGS 18: How this course is going to work

To avoid the common pitfalls of intro programming courses, we're going to take the following approach:

First 2/3 of course: basic concepts
Last 1/3: project-based; application

1. Pre-lecture videos + quiz (comprehension)
2. In-class practice (no stakes)
   - Google Forms for comprehension
3. Coding Labs (low stakes)
   - Staff/classmates there to help
4. Assignments (mid stakes)
   - Completed individually (*can* work together)
5. Exams + Final (higher stakes)
   - Closed notes
   - Exams: TTC-CBTF
   - Completed totally individually

Note: Google Forms are for *extra credit only*; every lecture attended is EC (up to 2%); can attend either lecture time on a given day

Your point of contact for COGS 18 will be the course website: https://cogs18.github.io

Canvas	https://canvas.ucsd.edu/courses/74370	grades, OH times/locations
Course Website	https://cogs18.github.io	Syllabus & links to stuff below
PrairieLearn	https://us.prairielearn.com/pl/course_instance/212605	Lecture notes + Assessments (Video quizzes, coding labs, assignments, exams) and regrade requests
Textbook (free)	https://shanellis.github.io/pythonbook/	
Piazza	https://piazza.com/ucsd/spring2026/cogs18 (see Canvas for sign-up code)	questions, discussion

Order I reply:

1. Piazza
2. Email
3. Canvas

In technical classes, Piazza is a particularly helpful resource

There are rules:

1. No duplicates.
2. Include Assignment & Question in Summary line.
3. Posts should include your question, what you've tried so far, and resources used.
4. Public posts are best.
5. Helping one another is encouraged.
6. No assignment code in public posts.
7. We're not robots.

COGS 18: How You'll Be Evaluated

	% of Grade	Requirement
Pre- and Post-Surveys	2%	2 surveys
Video Quizzes	12%	12 VQs
Coding Labs	16%	8 Coding Labs
Assignments	20%	5 assignments
Midterms	30%	2 midterms
Final	20%	Complete final exam or final project

The “I already know Python” option

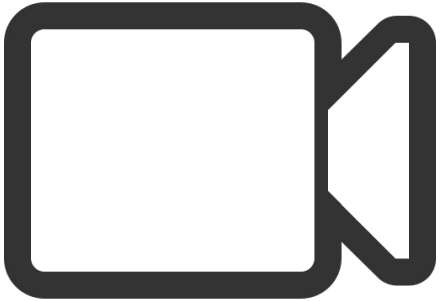
Graded on:

- Pre- and Post (2% each)
- (2) Midterm Exams (60%)
- Final Exam (38%)

Opt in:

- Via Google Form (link also on syllabus)
- Once you opt in you cannot change your mind
- If you are unsure, proceed with full course for now; you can always opt-in later.

Video Quizzes (12%)



The night before most lectures (Mon and Wed), you'll have ~3 videos (~3 min each) to watch from our textbook and a quiz to complete. Each quiz is worth 1% of your grade.

Why? Student feedback/experience. We're taking the most basic concepts out of lecture to allow for harder examples and more hands-on activities in-class

https://shanellis.github.io/pythonbook/content/03-basics/variables.html

Q1: VQ1: Variables

Total points: 0/1

0%

Assessment is **open** and you can answer questions.
Available credit: 100% (Staff override)

Question	Status	Available points	Awarded points
w/o LLM: Lecture Review: Syllabus Review			
Question 1	unanswered	0.05, 0.02 MC	— /0.05
Question 2	unanswered	0.05 TF	— /0.05
Question 3	unanswered	0.05, 0.04, 0.03, 0.02 CB	— /0.05
w/o LLM: Variable Assignment			
Question 4	unanswered	0.1, 0.1, 0.1, 0.08, 0.08, 0.08, 0.06, 0.06, 0.06, 0.04 Write code	— /0.1
Question 5	unanswered	0.1, 0.08, 0.06, 0.04	— /0.1
w/o LLM: Integers and Floats			
Question 6	unanswered	0.1, 0.05	— /0.1
Question 7	unanswered	0.05	— /0.05
Question 8	unanswered	0.1, 0.05	— /0.1
w/o LLM: Strings and Booleans			
Question 9	unanswered	0.1, 0.05	— /0.1
Question 10	unanswered	0.1, 0.1, 0.1, 0.08, 0.08, 0.08, 0.06, 0.06, 0.06, 0.04	— /0.1
Question 11	unanswered	0.1, 0.05	— /0.1
Question 12	unanswered	0.1	— /0.1

CodingLabs: apply concepts discussed in lecture using **coding labs (16%)**. Practice makes progress.

Unlimited attempts for full credit (2% each)

- Coding Labs will be completed on PrairieLearn
- Encouraged to work with others
- Unlimited attempts

You **should attend the section to which you're assigned**. You *can* attend a different section. However, if one section becomes too crowded each week, we'll revisit this policy.

<u>Note</u>: CodingLabs (and Office Hours) begin **Week 2**

(5) Assignments (20%) : After you've seen the material in VQ, lecture, and coding lab....

Assignments due Sun @ 11:59 PM.

Assignment	Week	Median Time Spent (hours)
A1	wk3	2
A2	wk4	4
A3	wk5	4
A4	wk7	5
A5	wk9	5

(2) Midterms (30%): will be completed individually.

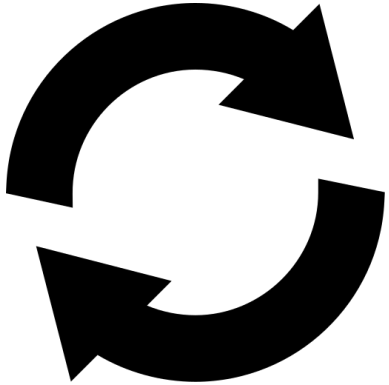
Midterms (E1 & E2)

- In-person, closed notes
- Focused on: concepts, code reading and debugging

These are completed at the TTC-CBTF.

Sign up for midterms asap! …but don't just pick the latest time slot!

You *can* sign up for a practice exam slot week 1/2 (3/30-4/10) (encouraged for: nervous test takers/anxious about new environments; NOT required)



Exam re-take (*optional*; weeks 6 + 9)

- Must have taken exam as originally scheduled
- Will take 75% of highest score and 25% of lowest score
- <u>You can't sign up for this yet.</u> (opens 1wk prior)

Final (30%): You've got options…

Final Exam

- Cumulative, in-person, closed notes exam at TTC-CBTF (45min)
- Take-home (untimed) guided, mini-project
- Estimated total time: 2h

Final Project

- Independent coding project to be completed by Sunday before finals week
- 15 min in-person oral exam (discussing your project!)
- Estimated total time: 6-10h

Any questions about course logistics?

Where to turn for help and practice when learning to program?

Many Avenues for help

- Office Hours
- Classmates
- Piazza
- The Internet
- GenAI (ChatGPT, Claude, CoPilot, etc.)

You ARE encouraged to help one another in COGS 18.

Discuss: GenAI

(ChatGPT, Claude, CoPilot, etc.)

First: tell the people around you your name

Then, discuss (one or all of):

1. How has GenAI been helpful to you in your learning generally? (Or, why have you not used it?)
2. How do you think it could be helpful in this course?
3. How do you think it could limit your learning? What do you want to avoid?

When CAN'T you use LLMs in COGS 18? Computer-Based **Exams**

When CAN you use them? **Everything else** (but we'll be clear about what you *should* know without an LLM; I'll demo use!)

…but you need to be able to explain everything you turn in.
Be able to answer yes to BOTH:

- Can I explain each piece of code and each analysis carried out in what I'm submitting?
- Could I reproduce this code on my own?

Process:

- Try on your own first
- Avoid copy + pasting whole questions
- Come up with your own questions
- Read *and think* about the answers
- Work to <u>understand</u>

Using AI/LLMs as a beginner

Why do anything w/o GenAI?

You don’t know what you don’t know

Core concepts necessary for prompting

Discuss: Google Maps crutch (looking things up forever; not actually saving time)

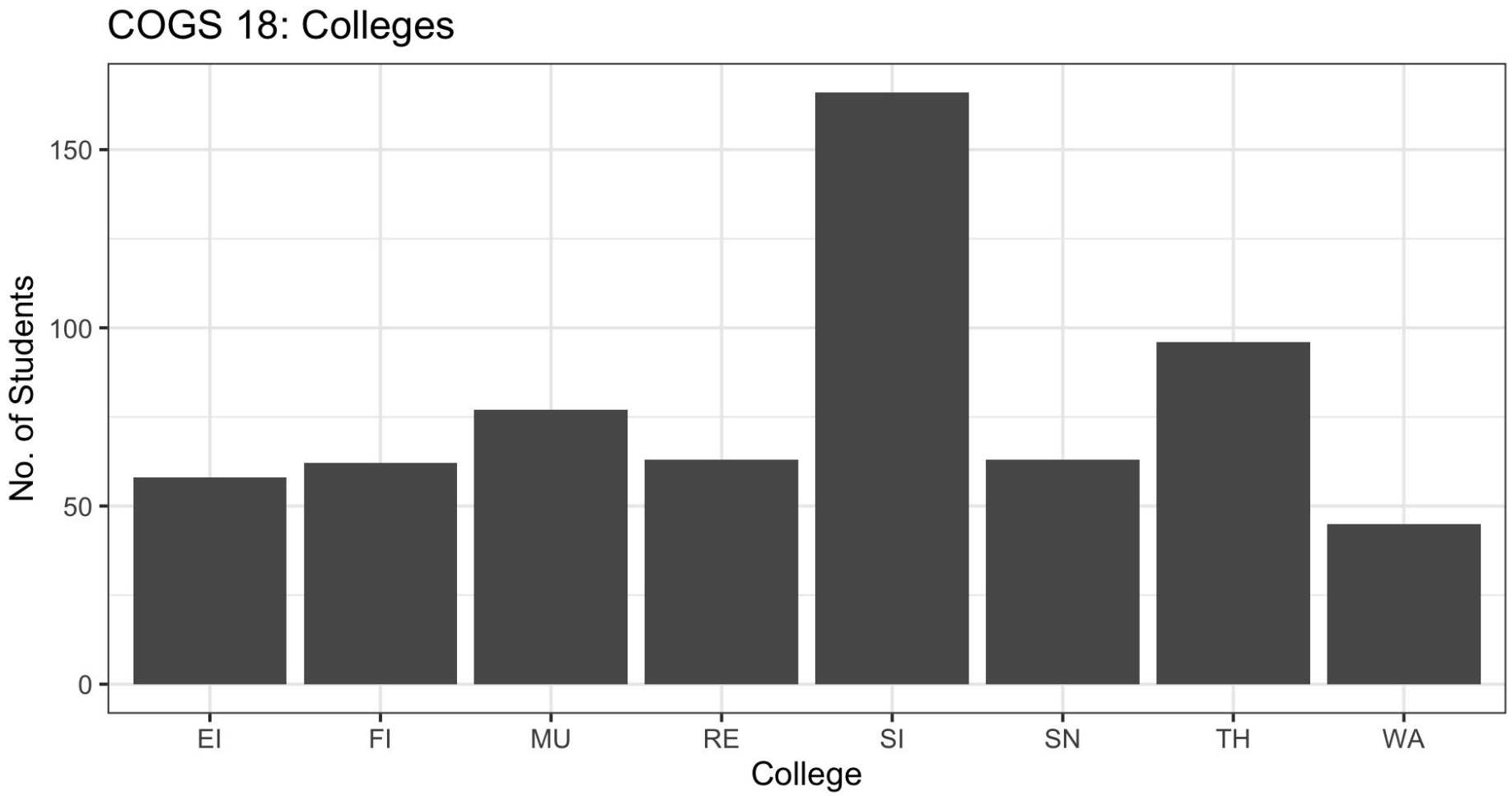
A message for first-gen students, transfer students, and those who don't have older siblings/friends who have attended UCSD/university

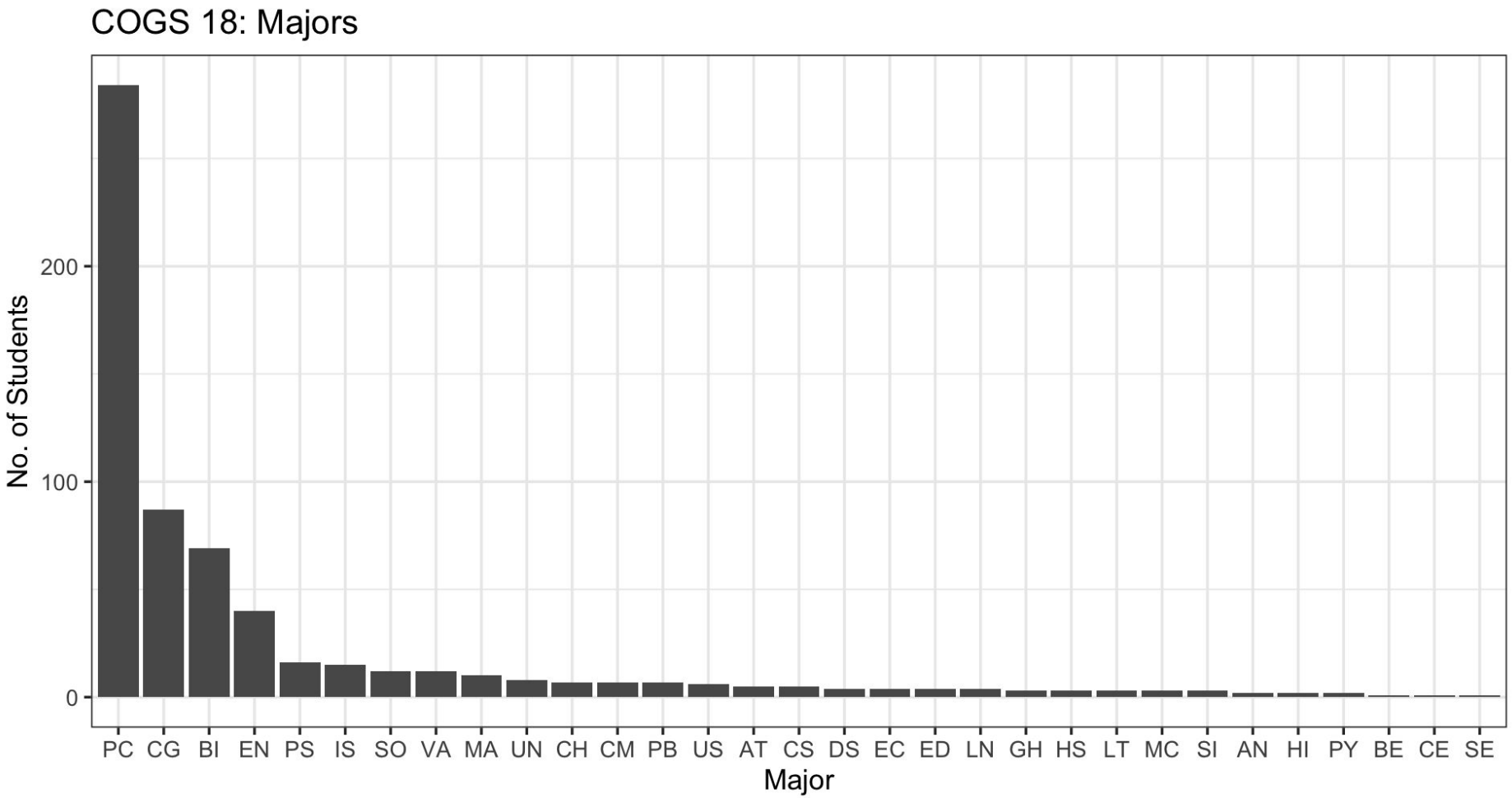
If you are struggling, come to office hours. Attend extra help sessions. Ask questions on Piazza. Reach out to me to ask for better approaches. Your classmates ARE doing this. And, you're not alone.

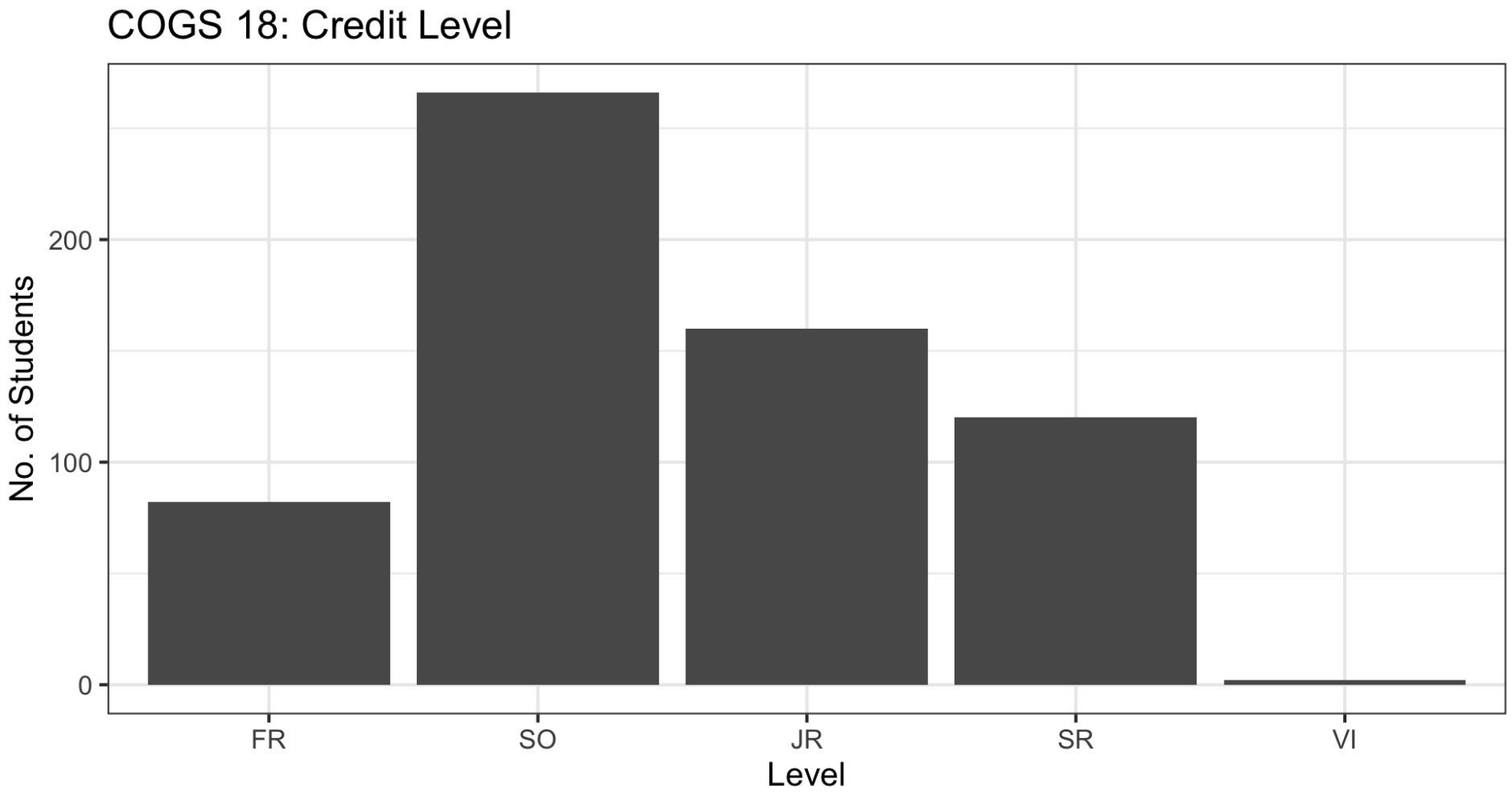
If you need a bit longer on something b/c you fell sick, a family thing came up, work called you in for an extra shift, etc., ask for an extension. Your classmates ARE doing this.

Today I used a PDF slideshow, but every other day of class, lecture notes will be presented in a Jupyter notebook

hello
my name is


COGS 18: Colleges
No. of Students
150
100
50
0
EI
FI
MU
RE
SI
SN
TH
WA
College


COGS 18: Majors
No. of Students
200
100
0
PC CG BI EN PS IS SO VA MA UN CH CM PB US AT CS DS EC ED LN GH HS LT MC SI AN HI PY BE CE SE
Major


COGS 18: Credit Level
No. of Students
200
100
0
FR
SO
JR
SR
VI
Level

Please complete the pre-course survey before Sunday after week 2.

Sign up for exam slots on PrairieTest.